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INSTITUT D'INVESTIGACIÓ EN INTELLIGÈNCIA ARTIFICIAL

CSIC
CONSEJO SUPERIOR DE INVESTIGACIONES CIENTÍFICAS

Integrating Face Generation on a (ROS) Robot

Vizij WS – HRI'26

March 2026

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What did we talk about today?

- gaze direction, 'pointing' with gaze
- gaze 'policy' (following a face, gaze aversion...)
- expression/'state of being' ('sick', 'hungry'...)
- extra features (tears,...)
- lip sync
- facial action units
- background colour, colour palette, 'warmth', texture
- gaze to orchestrate interaction
- planning gazing behaviour
- 'face as a canvas'
- ...

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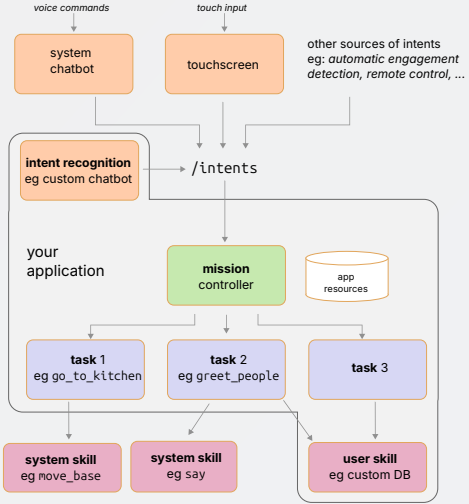
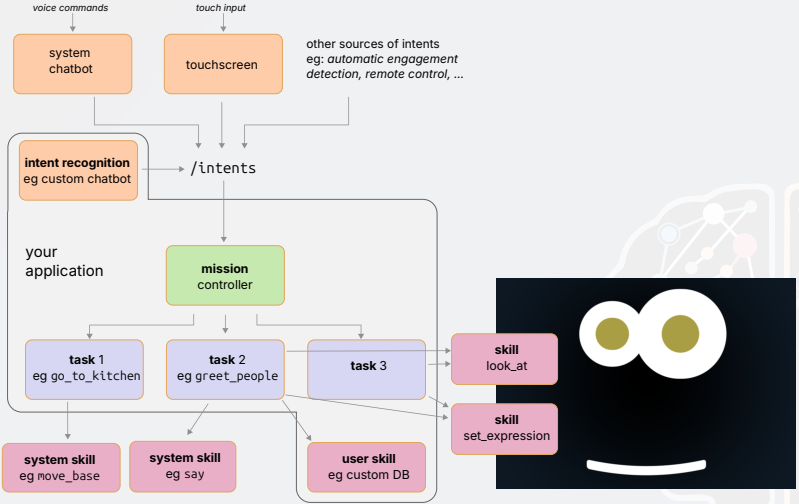
Architecture for interaction 101

Software architecture for interactive robots



Where does the face fit in?



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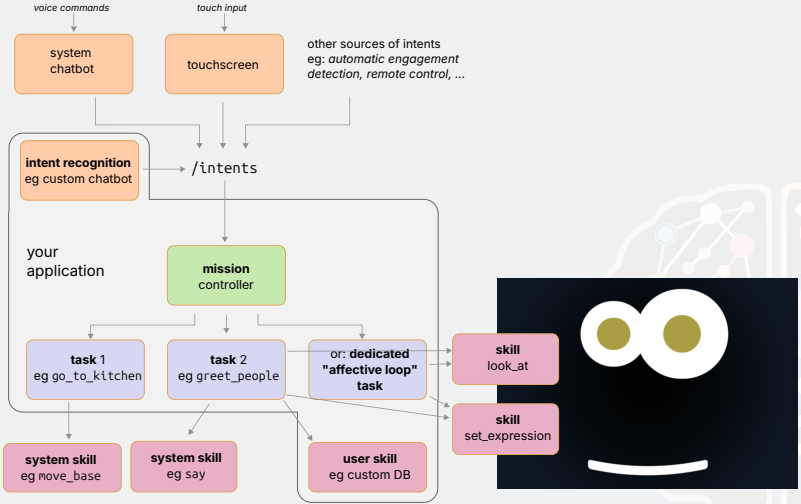

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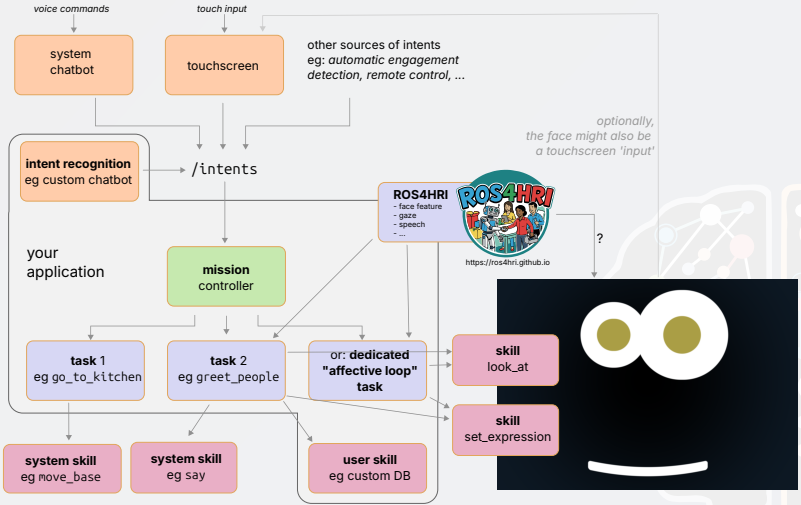

ROS API for faces

A ROS API for robot face

In:

- **input topics**
  - `/skill/set_expression` (`interaction_skills/SetExpression`) → **skill**
  - `/my_face/speech` (`std_msgs/String`)
  - `/my_face/closed_captions` (`hri_actions_msgs/ClosedCaption`)
- **action server**
  - `/skill/look_at` (`interaction_skills/LookAt`) → **skill**

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Out:

- **output topic**
  - `/my_face/image_raw` (`sensor_msgs/Image`)
  - **and/or** directly display the face in a window

One example: the `rpk` face template

Creating a Face with rpk

Generate a new ROS4HRI compliant face node in seconds using the rpk template generator:

```
$ pip3 install rpk
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$ pip3 install rpk
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$ pip3 install rpk
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$ colcon build && source install/setup.bash
```
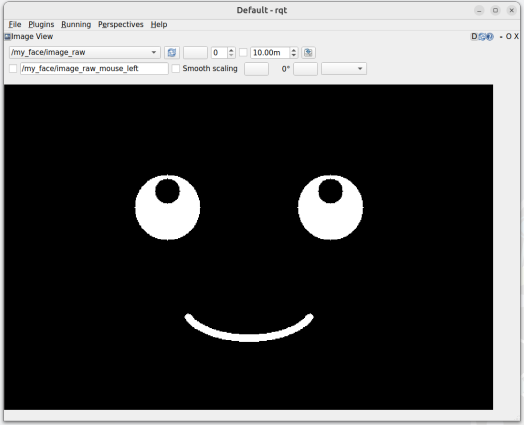
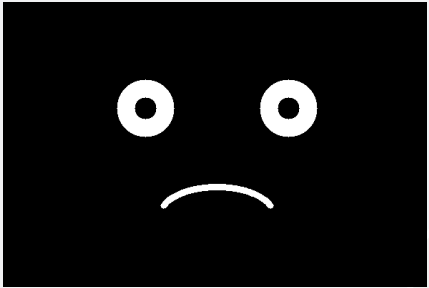

ROS API in action



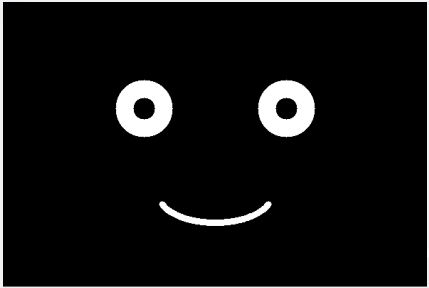

```
$ ros2 launch my_beautiful_face my_beautiful_face.launch.py
```

ROS API in action



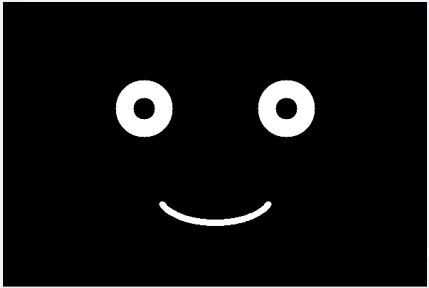
```
$ ros2 topic pub
            /skill/set_expression
            interaction_skills/msg/SetExpression "expression: {expression: sad}"
```

ROS API in action



```
$ ros2 topic pub
            /skill/set_expression
            interaction_skills/msg/SetExpression "expression: {expression: happy}"
```

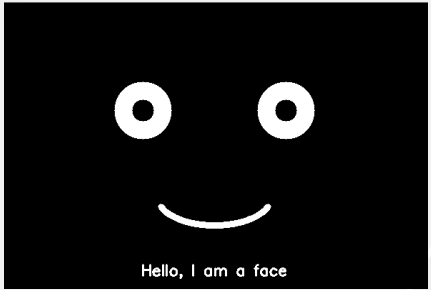
ROS API in action



```
$ ros2 topic pub
            /skill/set_expression
            interaction_skills/msg/SetExpression "expression: {expression: happy}"
```

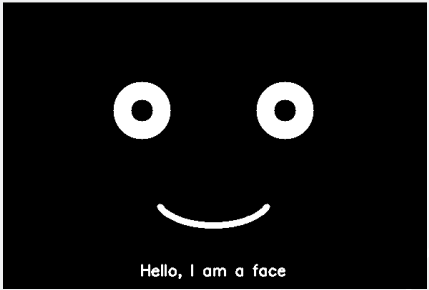
(the `SetExpression` also supports continous arousal/valence)

ROS API in action




```
$ ros2 topic pub
            /my_beautiful_face/closed_captions
           hri_actions_msgs/msg/ClosedCaption  "text: 'Hello, I am a face'"
```

ROS API in action

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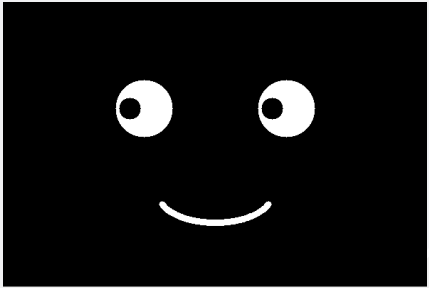
$ ros2 topic pub
        /my_beautiful_face/closed_captions
       hri_actions_msgs/msg/ClosedCaption  "text: 'Hello, I am a face'"
```

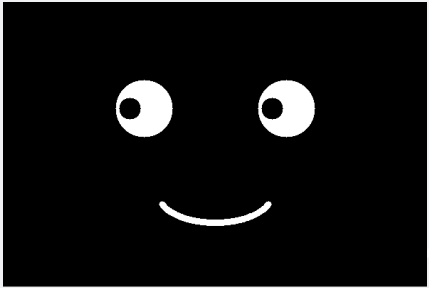
(use `/my_beautiful_face/speech` for the vismes)

ROS API in action



```
$ ros2 action send_goal
            /skill/look_at
            interaction_skills/action/LookAt "target: {point: {x: 0.0, y: 1.0, z: 1.0}}"
```

ROS API in action



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            interaction_skills/action/LookAt "target: {point: {x: 0.0, y: 1.0, z: 1.0}}"
```
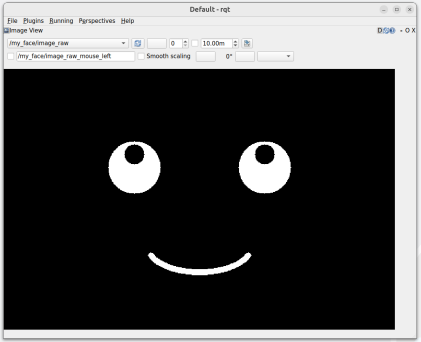
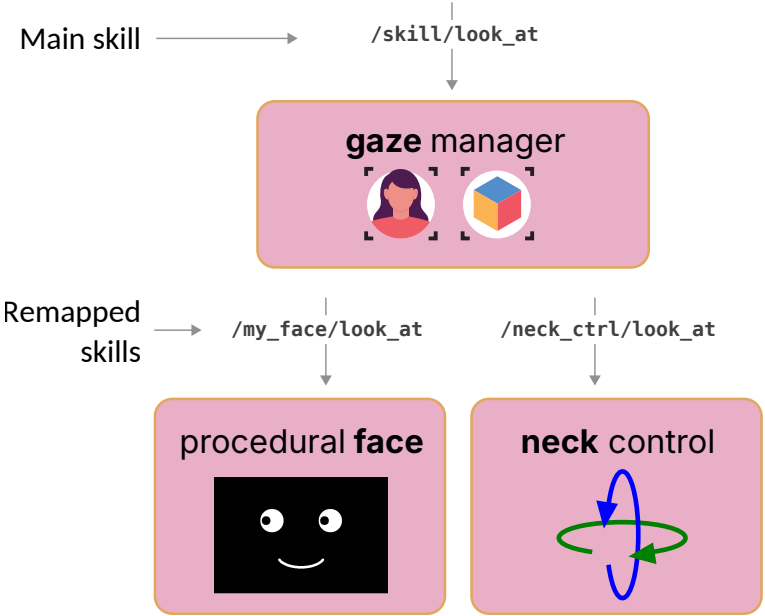

(the `LookAt` skill has more to it, including gazing policies)

ROS API in action

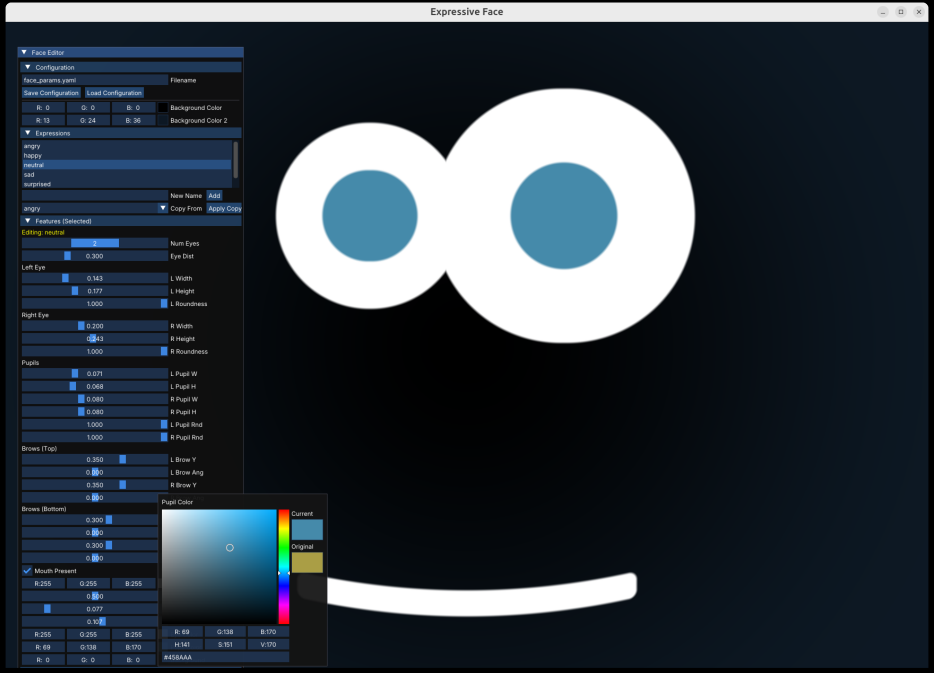


In the `rpk` template, the face is drawn manually, using `OpenCV`. Up to you to replace it with a more sophisticated face generator (e.g. vizij)!

A slightly more complex case to finish


Main skill
/skill/look_at
gaze manager
Remapped skills
/my_face/look_at
/neck_ctrl/look_at
procedural face
neck control

Happy face building!


Expressive Face
Face Editor
Configuration
face_params.yaml
Filename
Save Configuration
Load Configuration
R: 0
G: 0
B: 0
Background Color
R: 13
G: 24
B: 36
Background Color 2
Expressions
angry
happy
neutral
sad
surprised
New Name
Add
angry
Copy From
Apply Copy
Features [Selected]
Editing: neutral
2
Num Eyes
0.300
Eye Dist
Left Eye
0.143
L Width
0.177
L Height
1.000
L Roundness
Right Eye
0.200
R Width
0.243
R Height
1.000
R Roundness
Pupils
0.071
L Pupil W
0.068
L Pupil H
0.080
R Pupil W
0.080
R Pupil H
1.000
L Pupil Rnd
1.000
R Pupil Rnd
Brows (Top)
0.350
L Brow Y
0.000
L Brow Ang
0.350
R Brow Y
0.000
Brows (Bottom)
0.300
0.000
0.300
0.000
Mouth Present
R:255
G:255
B:255
0.000
0.077
0.107
R:255
G:255
B:255
R: 69
G:138
B:170
R: 0
G: 0
B: 0
Pupil Color
Current
Original
R: 69
G:138
B:170
H:141
S:151
V:170
#458AAA